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Logical Foundations of Quantum Mechanics

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I declare that the generative artificial intelligence tool ChatGPT-5.4 was used during the preparation of this thesis for formal and linguistic assistance, namely for generating and adjusting LaTeX code, and grammar correction and editing.

I revised and checked all suggested outputs and take full responsibility for the final form and content of the thesis.

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Title: Logical Foundations of Quantum Mechanics

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Abstract: This thesis studies residuated ortholattices (ROL) and their associative subvariety (A-ROL) in the context of quantum logic. Residuated ortholattices provide a broader algebraic setting for orthomodular lattices by requiring residuation of the Sasaki product. The thesis includes an explicit equational proof that ROL is a variety and situates both ROL and A-ROL within the surrounding lattice-theoretic landscape. In particular, it proves that ROL properly extends OML, while A-ROL is a proper subvariety of ROL incomparable with OML. It is further shown that, in every algebra in A-ROL, the Sasaki-product reduct is a left regular band, whereas every proper equational strengthening in the pure language of the Sasaki product collapses to Boolean algebras. Finally, the thesis introduces and analyzes an explicit family of finite algebras to prove that A-ROL has infinitely many subvarieties.

Keywords: residuated ortholattices, associative residuated ortholattices, quantum logic, Sasaki product, orthomodular lattices

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Abstrakt: Tato práce se zabývá reziduovanými ortosvazy (ROL) a jejich asociativní podvarietou (A-ROL) v kontextu kvantové logiky. Reziduované ortosvazy poskytují širší algebraický rámec pro ortomodulární svazy tím, že vyžadují reziduaci Sasakiho součinu. Práce obsahuje důkaz pomocí identit, že ROL tvoří varietu, a zasazuje tak ROL i A-ROL do okolního svazově-teoretického prostředí. Konkrétně dokazuje, že ROL je vlastní nadvarietou OML, zatímco A-ROL je vlastní podvarietou ROL, a že A-ROL je s OML neporovnatelná vzhledem k inkluzi. Dále je ukázáno, že pro každou algebru z A-ROL tvoří její redukt podle Sasakiho součinu levý regulární band, zatímco každé vlastní zesílení pomocí identity v čistém jazyce Sasakiho součinu vede ke kolapsu na Booleovy algebry. Nakonec práce konstruuje a analyzuje explicitní posloupnost konečných algeber, pomocí níž dokazuje, že A-ROL má nekonečně mnoho podvariet.

Klíčová slova: reziduované ortosvazy, asociativní reziduované ortosvazy, kvantová logika, Sasakiho součin, ortomodulární svazy

Contents

Introduction

Quantum logic arose from the attempt to understand the algebraic structure of propositions in quantum mechanics. In the classical setting, propositions form a Boolean algebra, whereas in the quantum setting the Hilbert-space formalism leads instead to lattice structures of closed subspaces that are generally non-distributive. This failure of distributivity is one of the characteristic algebraic features of quantum logic. For the classical Birkhoff–von Neumann perspective, see [1], and for a broader overview of quantum logics, see [2].

This motivates the study of ortholattices and, in particular, orthomodular lattices. Orthomodular lattices provide the standard algebraic semantics for the best-known propositional quantum logics. At the same time, their algebraic theory has several well-known difficulties, concerning for example completions, proof theory, and decidability. In particular, it remains open whether the equational theory of orthomodular lattices is decidable. For recent discussion of these difficulties and of the motivation for moving beyond the orthomodular setting, see [3, pp. 37–38].

One way to study orthomodular behaviour in a broader algebraic setting is via residuation of the Sasaki product. In any bounded involutive lattice one defines

$$x * y := x \wedge (x' \vee y).$$

In orthomodular lattices this operation is residuated by the Sasaki hook. This motivates the class of *residuated ortholattices* introduced by Fussner and St. John. These algebras generalize orthomodular lattices and provide a wider variety in which orthomodular structures can be studied by translation and embedding methods analogous to negative translations from classical logic into intuitionistic logic. Moreover, ROL forms a variety and gives the equivalent algebraic semantics of an algebraizable propositional logic [3].

The present thesis studies residuated ortholattices and especially their associative subvariety. Since the Sasaki product is not associative in general, it is natural to consider the subvariety of *associative residuated ortholattices*, denoted A-ROL.

The thesis has two main aims. The first is to define and situate the classes ROL and A-ROL within the surrounding lattice-theoretic landscape and to analyze the structural consequences of associativity of the Sasaki product. The second is to prove that the variety A-ROL has infinitely many subvarieties.

The main results of the thesis are the following:

- The thesis gives an explicit equational proof that ROL is a variety.
- A-ROL is a proper subvariety of ROL.
- A-ROL and OML are incomparable under inclusion.
- For every algebra in A-ROL, the Sasaki-product reduct is a left regular band, and the first nontrivial strengthening in the pure language $\{*\}$ already forces collapse to Boolean algebras.
- There exists an explicit sequence of finite associative residuated ortholattices witnessing that A-ROL has infinitely many subvarieties.

Chapter 2 is primarily expository and summarizes results from Fussner and St. John. Chapters 3–5 contain the main results of the thesis along with some standard background facts.

The thesis is organized as follows:

- Chapter 1 collects the necessary preliminaries from lattice theory and universal algebra.
- Chapter 2 summarizes the background results on the relation between OML and ROL introduced in [3].
- Chapter 3 situates ROL and A-ROL among the relevant lattice-based classes and proves the key inclusion and separation results.
- Chapter 4 studies the Sasaki product under associativity and shows that the first proper strengthening in the pure language $\{*\}$ collapses to Boolean algebras.
- Chapter 5 introduces the family $(W_n)_{n\geq 1}$ and uses it to prove that A-ROL has infinitely many subvarieties.
- The concluding chapter summarizes the results and indicates several directions for further work.

1 Preliminaries

This chapter collects the basic notions and notation used throughout the thesis. We begin with a brief universal-algebraic background, then recall the lattice-theoretic concepts needed for ortholattices and orthomodular lattices, and finally introduce Sasaki operations, residuation, and the varieties of residuated ortholattices studied later.

1.1 Universal-algebraic background

We follow standard terminology from universal algebra [4].

Definition 1.1. An *algebra* is a structure

$$\mathbf{A} = (A, F),$$

where $A \neq \emptyset$ is a non-empty set and F is a family of finitary operations on A. The set A is called the *domain* of $\mathbf{A}$.

Definition 1.2. A *language* (or *type*) is a set of operation symbols, each equipped with a given arity. Algebras are said to be of the same language if they interpret the same operation symbols with the same arities.

Definition 1.3. An *identity* is a formal expression

$$s \approx t,$$

where s and t are terms of the same language. An algebra $\mathbf{A}$ *satisfies* the identity $s \approx t$, written

$$\mathbf{A} \models s \approx t,$$

if the interpretations of s and t in $\mathbf{A}$ coincide for every assignment of the variables.

Definition 1.4. A class $\mathcal{V}$ of algebras of the same language is called a *variety* if there exists a set Σ of identities such that

$$\mathcal{V} = \{\mathbf{A} : \mathbf{A} \models \sigma \text{ for all } \sigma \in \Sigma\}.$$

A *subvariety* of a variety $\mathcal{V}$ is a subclass of $\mathcal{V}$ that is itself a variety in the same language.

Definition 1.5. Let $\mathbf{A} = (A, F)$ and $\mathbf{B} = (B, F)$ be algebras of the same language. We say that $\mathbf{B}$ is a *subalgebra* of $\mathbf{A}$ if $B \subseteq A$ and B is closed under all operations of $\mathbf{A}$, that is, for every n-ary basic operation $f \in F$ and all $b_1, \ldots, b_n \in B$,

$$f^{\mathbf{A}}(b_1, \ldots, b_n) \in B.$$

Definition 1.6. Let $\mathbf{A} = (A, F)$ and $\mathbf{B} = (B, F)$ be algebras of the same language. A map $h\colon A \to B$ is called a *homomorphism* if for every n-ary basic operation $f \in F$ and all $a_1, \ldots, a_n \in A$,

$$h\Big(f^{\mathbf{A}}(a_1, \ldots, a_n)\Big) = f^{\mathbf{B}}\Big(h(a_1), \ldots, h(a_n)\Big).$$

If h is surjective, then $\mathbf{B}$ is called a *homomorphic image* of $\mathbf{A}$.

Definition 1.7. Let $\mathbf{A}$ be an algebra of language τ, and let $\sigma \subseteq \tau$ be a smaller language obtained by forgetting some operation symbols. The *σ-reduct* of $\mathbf{A}$, denoted by $\mathbf{A}\restriction_\sigma$, is the algebra obtained from $\mathbf{A}$ by keeping the same domain and interpreting only the operations from σ.

Conversely, if $\mathbf{A}$ is an algebra of language σ and $\tau \supseteq \sigma$, then an algebra $\mathbf{B}$ of language τ is called an *expansion* of $\mathbf{A}$ if $\mathbf{B}\restriction_\sigma = \mathbf{A}$.

Definition 1.8. Let $\mathbf{A} = (A, F)$ be an algebra. An equivalence relation θ on A is called a *congruence* of $\mathbf{A}$ if it is compatible with every basic operation of $\mathbf{A}$, that is, for every n-ary basic operation $f \in F$ and all $a_1, \ldots, a_n, b_1, \ldots, b_n \in A$,

$$a_1 \,\theta\, b_1, \ \ldots, \ a_n \,\theta\, b_n \quad \Longrightarrow \quad f(a_1, \ldots, a_n) \,\theta\, f(b_1, \ldots, b_n).$$

Definition 1.9. Let $\mathbf{A}$ be an algebra. The set of all congruences of $\mathbf{A}$ is denoted by

$$\mathrm{Con}(\mathbf{A}).$$

Ordered by inclusion, $\mathrm{Con}(\mathbf{A})$ forms a lattice, called a *congruence lattice* of $\mathbf{A}$.

Definition 1.10. An algebra $\mathbf{A}$ is called *simple* if

$$\mathrm{Con}(\mathbf{A}) = \{\Delta_A, \nabla_A\},$$

where Δ_A is the identity congruence and $\nabla_A = A^2$ is the universal congruence.

Definition 1.11. An algebra $\mathbf{A}$ is called *subdirectly irreducible* if $\mathrm{Con}(\mathbf{A})$ has a least nontrivial element, that is, there exists

$$\mu \in \mathrm{Con}(\mathbf{A}) \setminus \{\Delta_A\}$$

such that

$$\mu \leq \theta \qquad \text{for every } \theta \in \mathrm{Con}(\mathbf{A}) \setminus \{\Delta_A\}.$$

Definition 1.12. A variety $\mathcal{V}$ is called *congruence-distributive* if for every algebra $\mathbf{A} \in \mathcal{V}$, the congruence lattice $\mathrm{Con}(\mathbf{A})$ is distributive.

Definition 1.13. Let I be a non-empty set. A family $F \subseteq \mathcal{P}(I)$ is called a *filter over I* if

1. $I \in F$,

2. whenever $X, Y \in F$, then $X \cap Y \in F$,

3. whenever $X \in F$ and $X \subseteq Y \subseteq I$, then $Y \in F$.

A filter U over I is an *ultrafilter over I* if it is maximal among proper filters on I.

1.2 Lattices and ortholattices

Definition 1.14. A *lattice* is an algebra

$$\mathbf{L} = (L, \wedge, \vee)$$

such that for all $x, y, z \in L$:

$$\begin{aligned} x \wedge y &= y \wedge x, & x \vee y &= y \vee x, \\ x \wedge (y \wedge z) &= (x \wedge y) \wedge z, & x \vee (y \vee z) &= (x \vee y) \vee z, \\ x \wedge (x \vee y) &= x, & x \vee (x \wedge y) &= x. \end{aligned}$$

Definition 1.15. Let $\mathbf{L} = (L, \wedge, \vee)$ be a lattice. We define a partial order $\leq$ on L by

$$x \leq y \quad \Longleftrightarrow \quad x \wedge y = x.$$

Equivalently,

$$x \leq y \quad \Longleftrightarrow \quad x \vee y = y.$$

Definition 1.16. A *bounded lattice* is an algebra

$$\mathbf{L} = (L, \wedge, \vee, 0, 1)$$

such that $(L, \wedge, \vee)$ is a lattice and for all $x \in L$,

$$0 \leq x \leq 1.$$

Definition 1.17. Let $\mathbf{L} = (L, \wedge, \vee, 0, 1)$ be a bounded lattice. A unary operation $'$ on L is called an *involution* if for all $x, y \in L$:

$$\begin{gathered} x'' = x, \\ x \leq y \Rightarrow y' \leq x'. \end{gathered}$$

A bounded lattice equipped with an antitone involution is called a *bounded involutive lattice.*

Definition 1.18. An *ortholattice* is a bounded involutive lattice

$$\mathbf{L} = (L, \wedge, \vee, ', 0, 1)$$

such that for all $x \in L$:

$$\begin{gathered} x \wedge x' = 0, \\ x \vee x' = 1. \end{gathered}$$

Definition 1.19. An *orthomodular lattice* is an ortholattice

$$\mathbf{L} = (L, \wedge, \vee, ', 0, 1)$$

such that for all $x, y \in L$,

$$x \leq y \;\Rightarrow\; y = x \vee (y \wedge x'),$$

or equivalently in the equational form of orthomodularity:

$$x \vee (x' \wedge (x \vee y)) = x \vee y.$$

Definition 1.20. A lattice $\mathbf{L} = (L, \wedge, \vee)$ is called *modular* if for all $x, y, z \in L$,

$$((z \wedge y) \vee x) \wedge y = (x \wedge y) \vee (z \wedge y).$$

Definition 1.21. A lattice $\mathbf{L} = (L, \wedge, \vee)$ is called *distributive* if for all $x, y, z \in L$,

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z).$$

Definition 1.22. A *Boolean algebra* is a distributive ortholattice.

1.3 Sasaki operations and residuation

We follow the standard terminology for Sasaki product, Sasaki hook, and residuated ortholattices from [3].

Definition 1.23. Let $\mathbf{L} = (L, \wedge, \vee, ', 0, 1)$ be a bounded involutive lattice. The *Sasaki product* on L is the binary operation defined by

$$x * y := x \wedge (x' \vee y).$$

Definition 1.24. Let $\mathbf{L} = (L, \wedge, \vee, ', 0, 1)$ be a bounded involutive lattice. The *Sasaki hook* on L is the binary operation defined by

$$x \to y := x' \vee (x \wedge y).$$

Definition 1.25. Let $(P, \leq, *)$ be a partially ordered algebra with a binary operation $*$. A binary operation $\backslash$ on P is called a *right residual* of $*$ if for all $x, y, z \in P$,

$$x * y \leq z \quad \Longleftrightarrow \quad y \leq x \backslash z.$$

Remark 1.26. Equivalently, a binary operation $\backslash$ is a right residual of $*$ if for all $x, z \in P$, the element $x \backslash z$ is the greatest element of the set

$$\{\, u \in P : x * u \leq z \,\}.$$

Indeed,

$$x * u \leq z \iff u \leq x \backslash z.$$

Definition 1.27. A *residuated ortholattice*, briefly a *ROL*, is an algebra

$$\mathbf{A} = (A, \wedge, \vee, ', \backslash, 0, 1)$$

such that

1. $(A, \wedge, \vee, ', 0, 1)$ is an ortholattice, and
2. for all $x, y, z \in A$,
$$x * y \leq z \quad \Longleftrightarrow \quad y \leq x \backslash z.$$

Definition 1.28. An *associative residuated ortholattice*, briefly an *A-ROL*, is a residuated ortholattice

$$\mathbf{A} = (A, \wedge, \vee, ', \backslash, 0, 1)$$

such that its Sasaki product is associative, that is, for all $x, y, z \in A$,

$$x * (y * z) = (x * y) * z.$$

Lemma 1.29. *Let* $\mathbf{L} = (L, \wedge, \vee, ', 0, 1)$ *be a bounded involutive lattice. Then for all* $x, y \in L$*,*

$$
\begin{aligned}
x * x &= x, \\
x * 1 &= x, \\
1 * x &= x, \\
0 * x &= 0, \\
x * 0 &= x \wedge x', \\
x \wedge y \leq x &* y \leq x.
\end{aligned}
$$

Proof. These are standard properties of the Sasaki product, see [3, Lemma 2.3(2)–(5)]. For completeness, let us verify them directly.

By definition,

$$x * y = x \wedge (x' \vee y).$$

Substituting $y := x$, we obtain

$$x * x = x \wedge (x' \vee x) = x \wedge 1 = x.$$

Similarly,

$$x * 1 = x \wedge (x' \vee 1) = x \wedge 1 = x,$$

and

$$1 * x = 1 \wedge (1' \vee x) = 1 \wedge (0 \vee x) = x.$$

Also,

$$0 * x = 0 \wedge (0' \vee x) = 0 \wedge (1 \vee x) = 0,$$

and

$$x * 0 = x \wedge (x' \vee 0) = x \wedge x'.$$

Finally, since $y \leq x' \vee y$, we have

$$x \wedge y \leq x \wedge (x' \vee y) = x * y,$$

while trivially

$$x * y = x \wedge (x' \vee y) \leq x.$$

□

1.4 Subclass relations

To navigate inclusion relations between algebraic classes, we fix the following terminology.

Definition 1.30. Let $\mathcal{K}$ and $\mathcal{L}$ be classes of algebras of the same language. We say that $\mathcal{K}$ is a *subclass* of $\mathcal{L}$ if

$$\mathcal{K} \subseteq \mathcal{L}.$$

If in addition $\mathcal{K} \neq \mathcal{L}$, we say that $\mathcal{K}$ is a *proper subclass* of $\mathcal{L}$ and write

$$\mathcal{K} \subsetneq \mathcal{L}.$$

Definition 1.31. Two classes $\mathcal{K}$ and $\mathcal{L}$ of algebras of the same language are said to be *incomparable* if

$$\mathcal{K} \nsubseteq \mathcal{L} \qquad \text{and} \qquad \mathcal{L} \nsubseteq \mathcal{K}.$$

2 Residuated Ortholattices as an Environment for Orthomodular Lattices

This chapter is expository. We summarize the relevant results from Fussner–St. John [3], which serve as background for the original results of Chapters 4 and 5. The aim is to isolate those structural facts that motivate the later focus on associativity of the Sasaki product and on the varietal behaviour of A-ROL. For further discussion of the motivations behind residuated ortholattices and of why ROL provides a suitable ambient framework for orthomodular lattices and quantum logic, see [3].

2.1 From Orthomodularity to Residuation

As recalled in the preliminaries, every bounded involutive lattice admits the Sasaki product

$$x * y := x \wedge (x' \vee y)$$

and the Sasaki hook

$$x \to y := x' \vee (x \wedge y).$$

As shown in [3, Proposition 2.8], in Orthomodular lattices, the Sasaki hook is in fact the right residual of the Sasaki product (which is not the case in general residuated ortholattices). Thus, if A is an orthomodular lattice, then for all $x, y, z \in A$,

$$x * y \leq z \quad \iff \quad y \leq x \to z.$$

This observation suggests that one may shift attention away from the orthomodular law itself and instead isolate residuation of the Sasaki product as a primitive algebraic property. In this way, orthomodular lattices may be viewed inside a broader class of ortholattice-based structures in which the Sasaki product remains residuated even when orthomodularity fails.

2.2 Orthomodular Lattices inside ROL

Every orthomodular lattice admits a canonical expansion to the language of residuated ortholattices by interpreting the residual through the Sasaki hook:

$$x \backslash y := x' \vee (x \wedge y).$$

With this definition, every orthomodular lattice becomes a residuated ortholattice (see [3, Proposition 2.8]).

Thus ROL may be regarded as a genuine extension of the orthomodular setting.

Also, orthomodular lattices can be characterized internally among residuated ortholattices. Namely, a residuated ortholattice is orthomodular if and only if its residual is already term-definable in the ortholattice language, equivalently, if and only if the identity

$$x\backslash y \approx x' \vee (x \wedge y)$$

holds. In this sense, orthomodular lattices are exactly those residuated ortholattices in which no genuinely new residual operation is needed (see [3, Theorem 2.7 and Proposition 2.8]).

This shows that the passage from OML to ROL is an extension obtained by allowing a residual operation that needs no longer collapse to the Sasaki hook.

2.3 The Orthomodular Skeleton

Another reason why residuated ortholattices provide a useful environment for orthomodular lattices is that every residuated ortholattice has an internal orthomodular part.

Definition 2.1. Let

$$\mathbf{A} = (A, \wedge, \vee, ', \backslash, 0, 1)$$

be a residuated ortholattice. Define a unary operation $\sim$ on A by

$$\sim x := x\backslash 0.$$

The operation $\sim$ plays the role of a second negation-like operation, distinct in general from the orthocomplement x'. Iterating it gives rise to the subset

$$A^{\sim\sim} := \{\sim\sim x : x \in A\}.$$

Theorem 2.2. *Let* $\mathbf{A}$ *be a residuated ortholattice. Then the structure induced on* $A^{\sim\sim}$ *is an orthomodular lattice.*

Proof. See [3, Lemma 4.5]. □

The orthomodular skeleton is one of the key conceptual links between ROL and OML. It shows that every residuated ortholattice contains a canonically determined orthomodular core obtained from the double application of the negation-like operation $\sim$. This supports the view that residuated ortholattices form an ambient variety for orthomodular behaviour in the same sense in which Heyting algebras provide an ambient setting for Boolean algebras via double negation.

This matters here because associativity of the Sasaki product is closely tied to the behaviour of this orthomodular core. In fact, recent work shows that for a residuated ortholattice A, associativity of the Sasaki product is equivalent to the condition that the orthomodular skeleton of A is Boolean.[5, Theorem 1]

In the next chapter, we will start investigating the classes ROL and A-ROL by placing them within the surrounding lattice-theoretic landscape and prove the main inclusion and separation results that will be needed later.

3 The Algebraic Landscape of ROL and $\mathrm{A\text{-}ROL}$

This chapter provides an algebraic map of the main classes relevant to the present thesis. We will situate residuated ortholattices and associative residuated ortholattices within the broader landscape of lattice-based structures.

The chapter is divided into two parts. First, we recall the classical lattice-theoretic background. We then move to the varieties of residuated ortholattices and associative residuated ortholattices.

3.1 Classical background

In this section we list the principal algebraic classes that will be used throughout the thesis and indicate their basic mutual position.

The classes most relevant for our purposes are the following:

- lattices (L),
- modular lattices (ML),
- distributive lattices (DL),
- ortholattices (OL),
- orthomodular lattices (OML),
- Boolean algebras (BA).

We begin with two schematic diagrams. The first records the classical lattice-theoretic inclusions. The second records the position of classes we are interested in, ROL and A-ROL.

In the remainder of this chapter, the relations in these diagrams will be justified by explicit inclusion and non-inclusion proofs.

First, we will recall some standard facts about relations between lattice varieties depicted Figure 3.1

Proposition 3.1. *Every Boolean algebra is an orthomodular lattice, and every orthomodular lattice is an ortholattice. Equivalently:*

$$\mathrm{BA} \subseteq \mathrm{OML} \subseteq \mathrm{OL}.$$

Proof. The inclusion $\mathrm{OML} \subseteq \mathrm{OL}$ follows from definitions, so it remains to prove

$$\mathrm{BA} \subseteq \mathrm{OML}.$$

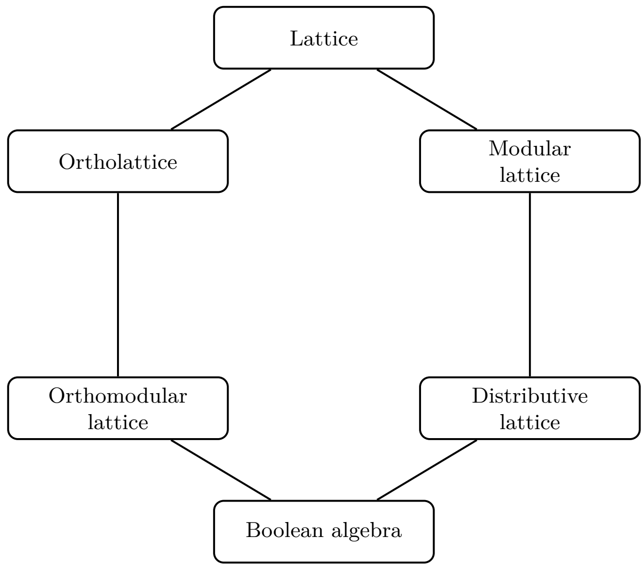


Figure 3.1 A Hasse diagram of the main classical lattice classes ordered by subclass inclusion.

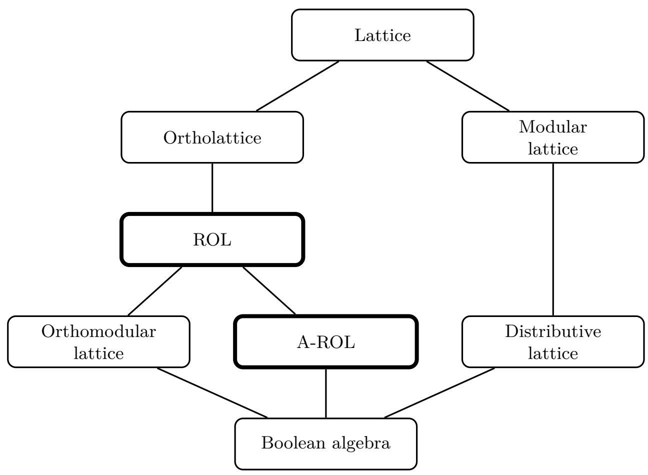


Figure 3.2 A Hasse diagram extending Figure 3.1 by adding varieties ROL and A-ROL.

Because every Boolean algebra is a distributive ortholattice by definition, we only need to show that BA proves orthomodularity.

$$\begin{aligned}
x \vee \big(x' \wedge (x \vee y)\big) &= (x \vee x') \wedge \big(x \vee (x \vee y)\big) && \text{by distributivity} \\
&= 1 \wedge \big(x \vee (x \vee y)\big) && \text{by } x \vee x' = 1 \\
&= 1 \wedge \big((x \vee x) \vee y\big) && \text{by associativity of } \vee \\
&= 1 \wedge (x \vee y) && \text{by idempotence of } \vee \\
&= x \vee y && \text{by greatest element.}
\end{aligned}$$

Hence BA satisfies the orthomodular law $x \vee (x' \wedge (x \vee y)) = x \vee y$, therefore

$$\mathrm{BA} \subseteq \mathrm{OML}.$$

□

We now consider the distributive side of the picture.

Proposition 3.2. *Every Boolean algebra is a distributive lattice, and every distributive lattice is a modular lattice, equivalently:*

$$\mathrm{BA} \subseteq \mathrm{DL} \subseteq \mathrm{ML} \subseteq \mathrm{L}$$

Proof. The inclusion BA $\subseteq$ DL again follows from definitions, so we only need to prove

$$\mathrm{DL} \subseteq \mathrm{ML}$$

$$\begin{aligned}
((z \wedge y) \vee x) \wedge y &= \big((z \wedge y) \wedge y\big) \vee (x \wedge y) && \text{by distributivity} \\
&= (z \wedge (y \wedge y)) \vee (x \wedge y) && \text{by associativity of } \wedge \\
&= (z \wedge y) \vee (x \wedge y) && \text{by idempotence of } \wedge \\
&= (x \wedge y) \vee (z \wedge y) && \text{by commutativity of } \vee .
\end{aligned}$$

Hence Modularity is provable in Distributive lattices and so every distributive lattice is also modular. □

Proposition 3.3. *An orthomodular lattice is Boolean if and only if it is distributive.*

Proof. Every Boolean algebra is by definition a distributive ortholattice, and also an orthomodular lattice by Proposition 3.1. Together this yields that every Boolean algebra is a distributive orthomodular lattice.

Conversely, let L be a distributive orthomodular lattice. Since an orthomodular lattice is an ortholattice, L is a distributive ortholattice. Therefore L is a Boolean algebra by definition. □

3.2 Residuated Ortholattices

We now pass from the classical setting to the framework of residuated ortholattices. From this point onward, the discussion turns to the main algebraic results developed in the thesis.

Residuated ortholattices provide a broader algebraic environment in which orthomodular behaviour can be studied.

Proposition 3.4. *The class* ROL *of residuated ortholattices is a variety.*

Proof. The fact that ROL is a variety is established in [3, pp. 39]. We include a proof by finding the explicit set of identities equivalent to residuation of sasaki product condition. By definition, a residuated ortholattice is an algebra

$$\mathbf{A} = (A, \wedge, \vee, ', \backslash, 0, 1)$$

such that $(A, \wedge, \vee, ', 0, 1)$ is an ortholattice and, if we define the Sasaki product by

$$x * y := x \wedge (x' \vee y),$$

then the following residuation condition holds for all $x, y, z \in A$:

$$x * y \leq z \iff y \leq x \backslash z.$$

We show that this condition is equivalent, over ortholattices, to the following finite set of identities:

$$(x * (x \backslash y)) \wedge y = x * (x \backslash y), \tag{R1}$$

$$y \wedge \Big(x \backslash (x * y)\Big) = y, \tag{R2}$$

$$x \backslash (y \wedge z) = (x \backslash y) \wedge (x \backslash z). \tag{R3}$$

Since the ortholattice axioms are equations, it will follow that ROL is equationally axiomatizable, hence a variety.

Let $\mathcal{K}$ be the class of all algebras

$$(A, \wedge, \vee, ', \backslash, 0, 1)$$

whose reduct $(A, \wedge, \vee, ', 0, 1)$ is an ortholattice and which satisfy (R1), (R2), and (R3). We prove that

$$\mathrm{ROL} = \mathcal{K}.$$

We first prove an auxiliary fact, that the Sasaki product is monotone in its second coordinate in ortholattices. If $u \leq v$, then by monotonicity of $\vee$,

$$x' \vee u \leq x' \vee v.$$

Applying monotonicity of $\wedge$, we obtain

$$x \wedge (x' \vee u) \leq x \wedge (x' \vee v).$$

Hence

$$\begin{aligned} x * u &= x \wedge (x' \vee u) \\ &\leq x \wedge (x' \vee v) \\ &= x * v. \end{aligned}$$

Let $\mathbf{A} \in \mathrm{ROL}$. We show that $\mathbf{A} \in \mathcal{K}$.

We verify (R1)–(R3) in ROL.

Verification of (R1). We must show

$$(x * (x\backslash y)) \wedge y = x * (x\backslash y).$$

Equivalently (in inequality form),

$$x * (x\backslash y) \leq y.$$

By residuation, this is equivalent to

$$x\backslash y \leq x\backslash y,$$

which is trivially true, hence (R1) holds.

Verification of (R2). Since

$$x * y \leq x * y$$

holds trivially, residuation yields

$$y \leq x\backslash(x * y).$$

Therefore

$$y \wedge \Big(x\backslash(x * y)\Big) = y,$$

which is (R2).

Verification of (R3). We prove both inequalities.

First, from (R1) with $y \wedge z$ in place of y, we get

$$x * (x\backslash(y \wedge z)) \leq y \wedge z.$$

Hence

$$x * (x\backslash(y \wedge z)) \leq y \qquad \text{and} \qquad x * (x\backslash(y \wedge z)) \leq z.$$

By residuation this gives

$$x\backslash(y \wedge z) \leq x\backslash y \qquad \text{and} \qquad x\backslash(y \wedge z) \leq x\backslash z.$$

Therefore the first inequality:

$$x\backslash(y \wedge z) \leq (x\backslash y) \wedge (x\backslash z).$$

For the second inequality, since

$$(x\backslash y) \wedge (x\backslash z) \leq x\backslash y,$$

monotonicity of $*$ in the second coordinate gives

$$x * \Big((x\backslash y) \wedge (x\backslash z)\Big) \leq x * (x\backslash y).$$

By (R1) we have $x * (x\backslash y) \leq y$, so

$$x * \Big((x\backslash y) \wedge (x\backslash z)\Big) \leq y.$$

Similarly,
$$x * \big((x\backslash y) \wedge (x\backslash z)\big) \leq z.$$
Hence
$$x * \big((x\backslash y) \wedge (x\backslash z)\big) \leq y \wedge z.$$
By residuation we then get the second inequality

$$(x\backslash y) \wedge (x\backslash z) \leq x\backslash(y \wedge z).$$

Together with previous inequality yielding

$$x\backslash(y \wedge z) = (x\backslash y) \wedge (x\backslash z),$$

proving (R3).

So $\mathbf{A} \in \mathcal{K}$ and hence $\mathbf{ROL} \subseteq \mathcal{K}$.

Conversely, let $\mathbf{A} \in \mathcal{K}$. We show that $\mathbf{A} \in \mathrm{ROL}$.

We now prove the residuation law

$$x * y \leq z \iff y \leq x\backslash z.$$

Left-to-right. Assume
$$x * y \leq z.$$
Then, by the definition of the lattice order,

$$(x * y) \wedge z = x * y.$$

By (R2) we have
$$y \wedge \big(x\backslash(x * y)\big) = y,$$
that is,
$$y \leq x\backslash(x * y).$$
Next, since we assume
$$(x * y) \wedge z = x * y,$$
we may substitute $(x * y) \wedge z$ for $x * y$ inside the term $x\backslash(\,*\,)$ and obtain

$$x\backslash(x * y) = x\backslash\big((x * y) \wedge z\big).$$

Now we use (R3) with $y := x * y$, which yields

$$x\backslash\big((x * y) \wedge z\big) = \big(x\backslash(x * y)\big) \wedge (x\backslash z).$$

Together with previous equation we get

$$x\backslash(x * y) = \big(x\backslash(x * y)\big) \wedge (x\backslash z).$$

By the definition of the lattice order again, this means that

$$x\backslash(x * y) \leq x\backslash z.$$

Combining this with
$$y \leq x \backslash (x * y),$$
we obtain
$$y \leq x \backslash z.$$
Right-to-left. Assume
$$y \leq x \backslash z.$$
Then
$$y \wedge (x \backslash z) = y.$$
Using monotonicity of $*$ in the second coordinate, we obtain
$$\begin{aligned} x * y &= x * \big(y \wedge (x \backslash z)\big) \\ &\leq x * (x \backslash z). \end{aligned}$$
By (R1) with $y := z$, we have
$$(x * (x \backslash z)) \wedge z = x * (x \backslash z),$$
so
$$x * (x \backslash z) \leq z.$$
Consequently,
$$x * y \leq z.$$
We have proved
$$x * y \leq z \iff y \leq x \backslash z,$$
so $\mathbf{A}$ is a residuated ortholattice. Hence $\mathbf{A} \in$ ROL and so $\mathcal{K} \subseteq \mathbf{ROL}$.

Thus ROL $= \mathcal{K}$. Since $\mathcal{K}$ is defined by a set of equations, the class ROL is a variety. □

Proposition 3.5.
$$\text{OML} \subseteq \text{ROL}.$$

Proof. Every orthomodular lattice admits a canonical expansion to a residuated ortholattice by taking the residual to be the Sasaki hook:
$$x \backslash y := x' \vee (x \wedge y).$$
Thus every orthomodular lattice gives rise to a residuated ortholattice. See [3, Proposition 2.8]. □

Remark 3.6. The inclusion is in fact proper. An explicit witness will be provided later by the algebra W_2 (see Figure 3.6, Chapter 5), which will be shown to belong to A-ROL $\subseteq$ ROL while failing to be orthomodular. Therefore
$$\text{OML} \subsetneq \text{ROL}.$$

3.3 Associative Residuated Ortholattices

We now isolate the associative part of the residuated ortholattices.

Definition 3.7. An *associative residuated ortholattice* is a residuated ortholattice in which the Sasaki product is associative. The corresponding class is denoted by A-ROL.

3.3.1 Inclusion in ROL

Proposition 3.8. *The class* A-ROL *is a proper subvariety of* ROL. *Equivalently,*

$$\text{A-ROL} \subsetneq \text{ROL}.$$

Proof. Since an associative residuated ortholattice is by definition a residuated ortholattice whose Sasaki product satisfies the identity

$$x * (y * z) \approx (x * y) * z,$$

the class A-ROL is obtained from ROL by adding one further identity. Hence A-ROL is a subvariety of ROL.

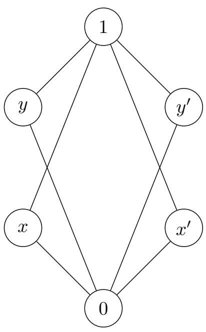
To see that the inclusion is proper, it suffices to find a residuated ortholattice whose Sasaki product is not associative, proving that

$$\text{A-ROL} \neq \text{ROL}.$$

To show the inequality, consider the six-element algebra

$$L = \{0, 1, x, x', y, y'\},$$

whose Hasse diagram and operation tables are displayed in Figure 3.3.



$'$	0	1	x	x'	y	y'
	1	0	x'	x	y'	y

$*$	0	1	x	x'	y	y'
0	0	0	0	0	0	0
1	0	1	x	x'	y	y'
x	0	x	x	0	x	x
x'	0	x'	0	x'	x'	x'
y	0	y	y	y	y	0
y'	0	y'	y'	y'	0	y'

$\backslash$	0	1	x	x'	y
0	1	1	1	1	1
1	0	1	x	x'	y
x	x'	1	1	x'	x'
x'	x	1	x	1	x
y	y'	1	y'	y'	1
y'	y	1	y	y	y

Figure 3.3 A six-element residuated ortholattice whose Sasaki product is not associative.

To show that $L \in \text{ROL}$, it remains to check the residuation law

$$a * b \le c \iff b \le a \backslash c \qquad \text{for all } a, b, c \in L.$$

Since L is finite and the operations $*$, $\backslash$, and the lattice order are given explicitly by the tables in Figure 3.3, this is a direct finite verification. Hence $L \in \text{ROL}$.

However, the Sasaki product on L is not associative. From the table for $*$ we obtain

$$x * y = x, \qquad y * y' = 0, \qquad x * y' = x, \qquad x * 0 = 0.$$

Therefore

$$(x * y) * y' = x * y' = x,$$

while

$$x * (y * y') = x * 0 = 0.$$

Hence

$$(x * y) * y' \neq x * (y * y'),$$

so $*$ is not associative on L. Thus $L \notin$ A-ROL.

Therefore

$$\text{A-ROL} \subsetneq \text{ROL}.$$

□

3.3.2 Incomparability with OML

The classes A-ROL and OML are not comparable under inclusion.

First we need to recall an important fact about orthomodular lattices. We fix the standard notation B_6 for the six-element ortholattice that plays an important role in the theory of orthomodular lattices. Its underlying set is

$$B_6 = \{0, 1, x, x', y, y'\},$$

with order given by

$$0 < y < x < 1, \qquad 0 < x' < y' < 1,$$

where 0 is the least element, 1 is the greatest element, x and x' are incomparable, y and y' are incomparable, and the orthocomplementation is given by

$$0' = 1, \qquad 1' = 0, \qquad x' = x', \qquad y' = y'.$$

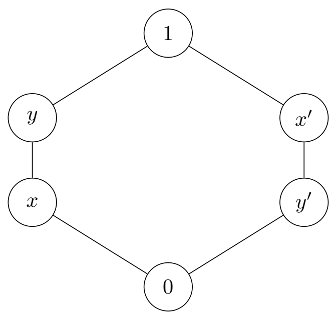


Figure 3.4 A Hasse diagram of a 6-element lattice B_6 (or benzene).

See Figure 3.4 for the Hasse diagram of such lattice.

Lemma 3.9. *Let L be an ortholattice. Then the following are equivalent:*

1. *L is orthomodular.*
2. *L does not contain B_6 as a subalgebra.*

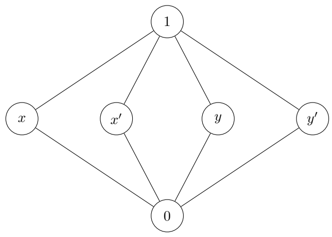
Proof. See [6, Chapter 1, Theorem 2 in the section "Orthomodular lattices"]. □

Proposition 3.10.

$$\mathrm{OML} \nsubseteq \text{A-ROL}.$$

Proof. Consider the 6-element ortholattice $L = \{0, 1, x, x', y, y'\}$ whose Hasse diagram is shown in Figure 3.5. The orthocomplementation is given by

$$x \leftrightarrow x', \qquad y \leftrightarrow y', \qquad 0 \leftrightarrow 1.$$



$'$	0	1	x	x'	y	y'
	1	0	x'	x	y'	y

$*$	0	1	x	x'	y	y'
0	0	0	0	0	0	0
1	0	1	x	x'	y	y'
x	0	x	x	0	x	x
x'	0	x'	0	x'	x'	x'
y	0	y	y	y	y	0
y'	0	y'	y'	y'	0	y'

$\backslash$	0	1	x	x'	y	y'
0	1	1	1	1	1	1
1	0	1	x	x'	y	y'
x	x'	1	1	x'	x'	x'
x'	x	1	x	1	x	x
y	y'	1	y'	y'	1	y'
y'	y	1	y	y	y	1

Figure 3.5 A Hasse diagram of a 6-element Orthomodular lattice in which Sasaki product is not associative.

We first show that L is orthomodular. By Lemma 3.9, it suffices to show that L does not contain B_6 as a subalgebra. Since L has exactly six elements, any subalgebra of L isomorphic to B_6 would necessarily have six elements as well, and hence would have to be all of L. Therefore, if L contained B_6 as a subalgebra, then necessarily $L \cong B_6$. But the Hasse diagram of L is not isomorphic to that of B_6: in L, all four nontrivial elements are incomparable to each other, whereas in B_6 the corresponding nontrivial elements are not arranged the same way. Hence $L \ncong B_6$, a contradiction.

Now define the Sasaki product on L by

$$a * b := a \wedge (a' \vee b).$$

We show that $*$ is not associative:

$$(x * y) * x' \neq x * (y * x'),$$

First,

$$x * y = x \wedge (x' \vee y) = x \wedge 1 = x,$$

since $x' \vee y = 1$. Therefore

$$(x * y) * x' = x * x' = x \wedge (x' \vee x') = x \wedge x' = 0.$$

On the other hand,

$$y * x' = y \wedge (y' \vee x') = y \wedge 1 = y,$$

since $y' \vee x' = 1$. Hence

$$x * (y * x') = x * y = x.$$

Thus

$$(x * y) * x' = 0 \neq x = x * (y * x'),$$

so the Sasaki product on L is not associative.

Therefore L is an orthomodular lattice whose Sasaki product is non-associative, and hence $L \notin$ A-ROL. □

Proposition 3.11. *The classes* A-ROL *and* OML *are incomparable under inclusion.*

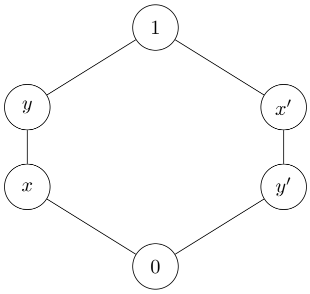
Proof. By Proposition 3.10, we already know that

$$\mathrm{OML} \not\subseteq \text{A-ROL}.$$

It therefore remains to show that

$$\text{A-ROL} \not\subseteq \mathrm{OML}.$$

Consider the 6-element algebra $L = \{0, 1, x, x', y, y'\}$ whose Hasse diagram and residuation and orthocomplement tables are shown in Figure 3.6.



$'$	0	1	x	x'	y	y'
	1	0	x'	x	y'	y

$\backslash$	0	1	x	x'	y	y'
0	1	1	1	1	1	1
1	0	1	x	x'	y	y'
x	x'	1	1	x'	1	x'
x'	y	1	y	1	y	y
y	x'	1	x'	x'	1	x'
y'	y	1	y	1	y	1

Figure 3.6 A Hasse diagram of a six-element associative residuated ortholattice that is not orthomodular, in this thesis we denote it W_2

The algebra displayed in Figure 3.6 is precisely the algebra W_2 that will be introduced in Chapter 5. By Theorem 5.6 (proven later), algebra W_2 belongs to A-ROL, hence

$$L \in \text{A-ROL}.$$

It remains to show that $L \notin \mathrm{OML}$. By Lemma 3.9, an ortholattice is orthomodular if and only if it does not contain B_6 as a subalgebra. But the ortholattice reduct of L is itself isomorphic to B_6. Therefore L is not orthomodular, and so

$$L \notin \mathrm{OML}.$$

Hence:

$$\text{A-ROL} \not\subseteq \mathrm{OML}.$$

□

3.3.3 Boolean Algebras inside A-ROL

Boolean algebras form a proper subclass of A-ROL.

Proposition 3.12.

$$\mathrm{BA} \subseteq \text{A-ROL}.$$

Proof. Let $A = (A, \wedge, \vee, ', 0, 1)$ be a Boolean algebra. Since every Boolean algebra is in particular an orthomodular lattice, A admits the canonical residual

$$x \backslash y := x' \vee (x \wedge y),$$

under which A becomes a residuated ortholattice.

It remains to show that the Sasaki product is associative. In a Boolean algebra, distributivity yields

$$x * y = x \wedge (x' \vee y) = (x \wedge x') \vee (x \wedge y) = 0 \vee (x \wedge y) = x \wedge y.$$

Thus in every Boolean algebra the Sasaki product coincides with the meet operation. Therefore, for all $x, y, z \in A$,

$$x * (y * z) = x \wedge (y \wedge z) = (x \wedge y) \wedge z = (x * y) * z.$$

Hence the Sasaki product is associative, and so $A \in$ A-ROL.

Therefore

$$\mathrm{BA} \subseteq \text{A-ROL}.$$

□

Proposition 3.13.

$$\text{A-ROL} \nsubseteq \mathrm{BA}.$$

Proof. By Proposition 3.11, there exists an algebra $L \in$ A-ROL such that

$$L \notin \mathrm{OML}.$$

Since every Boolean algebra is orthomodular, we have

$$\mathrm{BA} \subseteq \mathrm{OML}.$$

Therefore $L \notin$ BA. Hence

$$\text{A-ROL} \nsubseteq \mathrm{BA}.$$

□

Corollary 3.14. *The inclusion*

$$\mathrm{BA} \subsetneq \text{A-ROL}$$

is proper.

Proof. This follows immediately from Propositions 3.12 and 3.13. □

This chapter has located the main classes used in the thesis within a single algebraic picture. In particular, it has shown that ROL properly extends the orthomodular setting, that A-ROL is a proper subvariety of ROL, that A-ROL and OML are incomparable, and that BA is a proper subclass of A-ROL. These results clarify the position of associative residuated ortholattices within the surrounding landscape and motivate a closer study of the Sasaki product itself. The next chapter is devoted to that study.

4 Strengthenings of the Sasaki Product in A-ROL

This chapter studies what associativity of the Sasaki product contributes on the level of the pure multiplicative reduct. Our first goal is to identify the algebraic type of the reduct $(A, *)$ in associative residuated ortholattices. Our second goal is to understand how far one can strengthen this reduct by further identities in the language $\{*\}$ without forcing a collapse to Boolean algebras. This will connect the present theory with the theory of bands.

4.1 The Sasaki Product as a Left Regular Band Reduct

In this section we identify the basic band-theoretic behaviour of the Sasaki product in residuated ortholattices. We show that, already in every residuated ortholattice, the Sasaki product is idempotent and satisfies the left regular identity. Consequently, adding associativity yields a left regular band reduct.

Definition 4.1. A *band* is an algebra $(S, *)$ such that the binary operation $*$ is idempotent and associative, i.e. for all $x, y, z \in S$,

$$x * x = x \qquad \text{and} \qquad (x * y) * z = x * (y * z).$$

Definition 4.2. A band $(S, *)$ is called a *left regular band* if it satisfies

$$(x * y) * x = x * y \qquad \text{for all } x, y \in S.$$

The corresponding variety is denoted by LRB and its defining band axiom will be called lrb.

Recall that in a residuated ortholattice, the Sasaki product is defined by

$$x * y := x \wedge (x' \vee y).$$

Lemma 4.3. *In every residuated ortholattice, the Sasaki product is idempotent, i.e.*

$$x * x = x.$$

Proof. By definition of the Sasaki product,

$$x * x = x \wedge (x' \vee x).$$

Since $x' \vee x = 1$, we obtain

$$x * x = x \wedge 1 = x.$$

Hence the Sasaki product is idempotent. □

Lemma 4.4. *In every residuated ortholattice, the Sasaki product satisfies the left regular identity*

$$(x * y) * x = x * y.$$

Proof. The proof is adapted from [3, Lemma 2.5(2)].

By definition of $*$, we have

$$x * y = x \wedge (x' \vee y).$$

Therefore

$$(x * y) * x = \Big(x \wedge (x' \vee y)\Big) * x.$$

Expanding the product once more, we obtain

$$(x * y) * x = \Big(x \wedge (x' \vee y)\Big) \wedge \Big((x \wedge (x' \vee y))' \vee x\Big).$$

Using standard involutive lattice properties, this becomes

$$(x * y) * x = x \wedge (x' \vee y) \wedge \Big(x' \vee (x \wedge y') \vee x\Big).$$

Now by the absorption law,

$$x' \vee (x \wedge y') \vee x = x \vee x',$$

and hence

$$(x * y) * x = x \wedge (x' \vee y) \wedge (x \vee x').$$

Applying absorption once again, we get

$$(x * y) * x = x \wedge (x' \vee y).$$

Finally, by definition of $*$,

$$x \wedge (x' \vee y) = x * y.$$

Thus

$$(x * y) * x = x * y.$$

□

The previous two lemmas show that in every residuated ortholattice the operation $*$ already satisfies two of the three defining identities of a left regular band. Thus the only missing condition is associativity.

Corollary 4.5. *For every residuated ortholattice A, the following are equivalent:*

1. *A is an associative residuated ortholattice.*
2. *the reduct $(A, *)$ is a left regular band.*

Proof. By definition, an associative residuated ortholattice is exactly a residuated ortholattice in which $*$ is associative. By Lemmas 4.3 and 4.4, every residuated ortholattice already satisfies

$$x * x = x \qquad \text{and} \qquad (x * y) * x = x * y.$$

Hence, for a residuated ortholattice A, associativity of $*$ is equivalent to the condition that the reduct $(A, *)$ satisfies all defining identities of a left regular band. □

4.2 A Left Normal Strengthening and Boolean Collapse

By Corollary 4.5, once associativity is imposed, the Sasaki-product reduct of an associative residuated ortholattice is a left regular band. A natural next question is therefore whether further standard identities from band theory still admit non-Boolean subvarieties of A-ROL, or whether they force a collapse back to the classical Boolean setting.

One such strengthening is given by the identity

$$(z * x) * y = (z * y) * x,$$

which will later be shown to define the unique maximal proper subvariety below LRB in the lattice of band varieties.

Definition 4.6. A band $(S, *)$ is called *left normal* if it satisfies the identity

$$(z * x) * y = (z * y) * x \qquad \text{for all } x, y, z \in S.$$

For convenience, we denote this identity by **lr2** and the corresponding band variety is denoted by LR2.

The aim of this section is to show that, in the present setting, this strengthening is already too strong: once **lr2** is imposed on the Sasaki product in A-ROL, the resulting variety collapses to Boolean algebras. As we will prove later in this chapter, the **lr2** is in fact the first proper strengthening of the left regular band structure. The consequence of the theorem below is that the associative subvariety already lies at the edge of non-Boolean behaviour in the pure $*$ language: the first proper band-theoretic strengthening below LRB already forces Boolean collapse.

Theorem 4.7.

$$\text{A-ROL} + \{lr2\} = \text{BA}.$$

We prove this theorem in several steps. First, **lr2** forces commutativity of the Sasaki product.

Lemma 4.8. *In* A-ROL $+ \{lr2\}$*, the Sasaki product is commutative:*

$$x * y = y * x.$$

Proof. By **lr2**, we have

$$(x * y) * z = (x * z) * y \qquad \text{for all } x, y, z.$$

Substituting $x := 1$, and using

$$1 * u = 1 \wedge (1' \vee u) = 1 \wedge (0 \vee u) = u,$$

we obtain

$$y * z = z * y.$$

Hence $*$ is commutative. □

Next we show that, under **lr2**, the Sasaki product collapses to the meet operation.

Lemma 4.9. *In* A-ROL + $\{lr2\}$*, the Sasaki product coincides with meet:*

$$x * y = x \wedge y.$$

Proof. First we show that in every bounded involutive lattice, and hence in every residuated ortholattice, the following holds:

$$x \wedge y \leq x * y \leq x.$$

The first inequality

$$x \wedge y \leq x \wedge (x' \vee y) = x * y$$

holds since $y \leq x' \vee y$, and the second

$$x * y = x \wedge (x' \vee y) \leq x$$

holds by definition of meet.

By Lemma 4.8, $*$ is commutative, so also

$$x * y = y * x \leq y.$$

Hence

$$x * y \leq x \wedge y.$$

Combining this with $x \wedge y \leq x * y$, we conclude that

$$x * y = x \wedge y.$$

□

Once the Sasaki product is identified with meet, distributivity follows.

Lemma 4.10.

$$\text{A-ROL} + \{lr2\} \vdash x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z).$$

Proof. By Lemma 4.9, under the assumptions of A-ROL+$\{lr2\}$ the Sasaki product coincides with meet:

$$x * y = x \wedge y.$$

Hence the residuation law of residuated ortholattices becomes

$$x \wedge y \leq z \quad \Longleftrightarrow \quad y \leq x \backslash z.$$

We prove distributivity. First, in every lattice one always has

$$(x \wedge y) \vee (x \wedge z) \leq x \wedge (y \vee z),$$

since $x \wedge y \leq x$ and $x \wedge y \leq y \leq y \vee z$, and similarly $x \wedge z \leq x \wedge (y \vee z)$.

It remains to prove the converse inequality

$$x \wedge (y \vee z) \leq (x \wedge y) \vee (x \wedge z).$$

Let

$$a := (x \wedge y) \vee (x \wedge z).$$

Then clearly

$$x \wedge y \leq a \qquad \text{and} \qquad x \wedge z \leq a.$$

By residuation of $\wedge$, these inequalities imply

$$y \leq x\backslash a \qquad \text{and} \qquad z \leq x\backslash a.$$

Therefore

$$y \vee z \leq x\backslash a.$$

Applying residuation once more, we obtain

$$x \wedge (y \vee z) \leq a = (x \wedge y) \vee (x \wedge z).$$

Combining both inequalities, we get

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z).$$

□

Proof of Theorem 4.7. By Lemma 4.10, A-ROL with lr2 is distributive. Since every distributive ortholattice is a Boolean algebra, we conclude that

$$\text{A-ROL} + \{lr2\} \subseteq \text{BA}.$$

Conversely, every Boolean algebra is an orthomodular lattice, hence in particular a residuated ortholattice, and in every Boolean algebra expanded by the definition of the Sasaki product $x * y := x \wedge (x' \vee y)$, distributivity yields

$$x * y = x \wedge (x' \vee y) = (x \wedge x') \vee (x \wedge y) = 0 \vee (x \wedge y) = x \wedge y.$$

Hence the Sasaki product coincides with meet. Using this equation, we can then derive that Boolean algebras have associativity of the Sasaki product:

$$x * (y * z) = x \wedge (y \wedge z) = (x \wedge y) \wedge z = (x * y) * z,$$

and it satisfies **lr2**:

$$(z * x) * y = (z \wedge x) \wedge y = (z \wedge y) \wedge x = (z * y) * x.$$

Therefore every Boolean algebra satisfies the axioms of A-ROL $+ \{lr2\}$, and hence

$$\text{BA} \subseteq \text{A-ROL} + \{lr2\}.$$

Therefore,

$$\text{A-ROL} + \{lr2\} = \text{BA}.$$

□

Theorem 4.7 shows that the left normal strengthening lr2 is already too strong in the present setting: once lr2 is imposed on the Sasaki product in A-ROL, the resulting variety collapses to Boolean algebras.

In particular, A-ROL does not satisfy lr2. This may also be seen directly from the algebra W_2 introduced in the previous chapter as an associative residuated ortholattice that is not orthomodular (see Figure 3.6). In W_2 we have

$$1 * a_1 = a_1, \qquad 1 * a_2 = a_2, \qquad a_1 * a_2 = a_1, \qquad a_2 * a_1 = a_2.$$

Therefore, letting $z := 1$, $x := a_1$, and $y := a_2$, we obtain

$$(z * x) * y = (1 * a_1) * a_2 = a_1 * a_2 = a_1,$$

while

$$(z * y) * x = (1 * a_2) * a_1 = a_2 * a_1 = a_2.$$

Hence

$$(z * x) * y \neq (z * y) * x,$$

so lr2 fails in W_2.

4.3 Band-Theoretic Interpretation

The preceding results admit a natural interpretation in terms of band theory. By Corollary 4.5, the Sasaki-product reduct of every algebra in A-ROL is a left regular band. Thus, if we let

$$\mathcal{K} := \{\, (A, *) : A \in \text{A-ROL} \,\},$$

then

$$\mathcal{K} \subseteq \text{LRB}.$$

Moreover, by Theorem 4.7, the class A-ROL does not satisfy lr2. Hence $\mathcal{K}$ is not contained in LR2.

To analyze the exact band variety generated by these reducts, we now turn to the lattice of band varieties described by Fennemore [7], who determined the full lattice of varieties of bands. Figure 4.1 displays the lower fragment relevant for the present argument.

For clarity, the node corresponding to LRB is highlighted, the variety determined by lr2 and all varieties below it are shown as rounded rectangles, the varieties strictly above LRB are shown as rectangles, and the seven varieties incomparable with LRB are shown as ellipses.

For the present argument, the key point of Figure 4.1 is that the node corresponding to LRB has a unique maximal proper subvariety below it, namely LR2. Equivalently, every proper subvariety of LRB is contained in LR2.

Lemma 4.11. *Every proper subvariety of* LRB *is contained in* LR2.

Proof. This is read off from the relevant fragment of the lattice of varieties of bands in Figure 4.1, adapted from [7]. It follows from Fennemore's classification of varieties of bands (see [7, p. 174] for the classification result and [7, p. 176] for the lattice diagram). LR2 is the unique maximal proper subvariety immediately below LRB, so every proper subvariety of LRB must lie below LR2. □

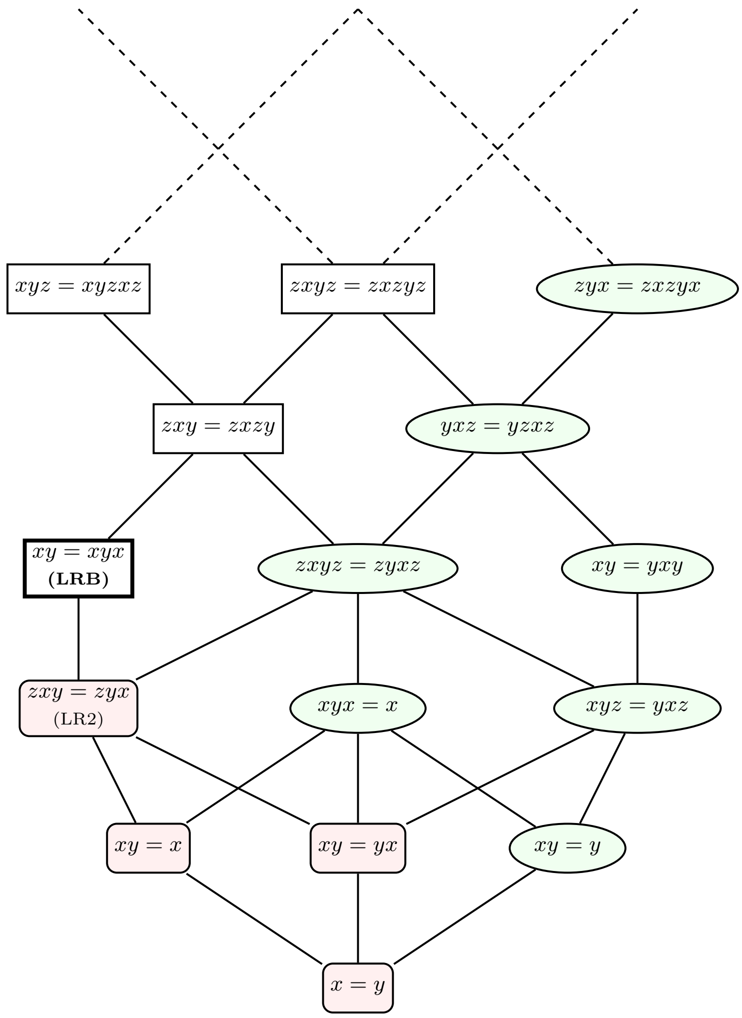


Figure 4.1 Lower fragment of the lattice of varieties of bands, adapted from [7, p. 176]. The node corresponding to LRB is highlighted.

Proposition 4.12. *Let*

$$\mathcal{K} := \{\, (A, *) : A \in \text{A-ROL} \,\}.$$

Then

$$V(\mathcal{K}) = \text{LRB}.$$

Proof. By Corollary 4.5, every member of $\mathcal{K}$ is a left regular band. Therefore

$$V(\mathcal{K}) \subseteq \text{LRB}.$$

On the other hand, A-ROL does not satisfy lr2, by Theorem 4.7. Hence there exists

$$A \in \text{A-ROL}$$

such that the reduct $(A, *)$ does not belong to LR2. It follows that

$$\mathcal{K} \nsubseteq \text{LR2},$$

and consequently

$$V(\mathcal{K}) \nsubseteq \text{LR2}.$$

Suppose, towards a contradiction, that

$$V(\mathcal{K}) \subsetneq \text{LRB}.$$

Then $V(\mathcal{K})$ is a proper subvariety of LRB. By Lemma 4.11, every proper subvariety of LRB is contained in LR2. Therefore

$$V(\mathcal{K}) \subseteq \text{LR2},$$

contrary to the previous paragraph. Hence

$$V(\mathcal{K}) = \text{LRB}.$$

□

Proposition 4.12 identifies the exact band variety generated by the Sasaki-product reducts of associative residuated ortholattices. Thus the pure $\{*\}$-equational behaviour of A-ROL is governed precisely by LRB, while Theorem 4.7 shows that imposing lr2 already forces collapse to Boolean algebras. We now combine these two facts.

Proposition 4.13. *Let ε be an equation in the language $\{*\}$. If*

$$\text{A-ROL} + \{\varepsilon\}$$

is a proper equational strengthening of A-ROL*, then*

$$\text{A-ROL} + \{\varepsilon\} \subseteq \text{BA}.$$

Proof. Assume that

$$\text{A-ROL} + \{\varepsilon\}$$

is a proper equational strengthening of A-ROL. Then not every algebra in A-ROL satisfies ε. Equivalently, not every algebra in $\mathcal{K}$ satisfies ε.

Since $V(\mathcal{K}) = \text{LRB}$ by Proposition 4.12, it follows that

$$\text{LRB} \not\models \varepsilon.$$

Hence

$$\text{LRB} + \{\varepsilon\}$$

is a proper subvariety of LRB. By Lemma 4.11, we obtain

$$\text{LRB} + \{\varepsilon\} \subseteq \text{LR2}.$$

Now let

$$A \in \text{A-ROL} + \{\varepsilon\}.$$

Then its $\{*\}$-reduct satisfies ε, and therefore belongs to $\text{LRB} + \{\varepsilon\} \subseteq \text{LR2}$. Thus A satisfies lr2, so

$$A \in \text{A-ROL} + \{\text{lr2}\}.$$

By Theorem 4.7,

$$\text{A-ROL} + \{\text{lr2}\} = \text{BA}.$$

Therefore $A \in \text{BA}$. Since A was arbitrary, we conclude that

$$\text{A-ROL} + \{\varepsilon\} \subseteq \text{BA}.$$

□

Corollary 4.14. *Among proper equational strengthenings of* A-ROL *by identities in the pure language* $\{*\}$ *of the Sasaki product,* A-ROL *is the last non-Boolean stage.*

Proof. Let ε be any equation in the language $\{*\}$ such that

$$\text{A-ROL} + \{\varepsilon\}$$

is a proper equational strengthening of A-ROL. Then Proposition 4.13 yields

$$\text{A-ROL} + \{\varepsilon\} \subseteq \text{BA}.$$

Thus every proper strengthening in the pure multiplicative language is already Boolean. □

Remark 4.15. Figure 4.1 also gives a convenient visual summary of the argument.

By Proposition 4.12, the relevant band variety is exactly LRB. Hence every proper equational strengthening of A-ROL corresponds to a proper subvariety of LRB, and therefore by Lemma 4.11 lies below LR2. In the diagram, these are precisely the rounded-rectangle nodes below the highlighted LRB-node. By Theorem 4.7, already the first such strengthening forces Boolean collapse.

The rectangular nodes above LRB do not yield genuine strengthenings, since they impose weaker identities than those already valid in LRB. The elliptical nodes incomparable with LRB become relevant only through their meet with LRB, which is again a proper subvariety of LRB, hence lies below LR2.

5 Infinitely Many Subvarieties of A-ROL

In this chapter we prove that the variety A-ROL has infinitely many subvarieties.

The proof proceeds by constructing a family of finite algebras that are simple enough to be analyzed explicitly, but sufficiently different from one another to generate new subvarieties. The role of the sequence $(W_n)_{n\geq 1}$ is precisely to witness this increasing complexity inside A-ROL. Their underlying ortholattice shape is very transparent: two finite chains joined only by 0 and 1, with the involution reversing the two branches. Yet this simple structure turns out to be strong enough to produce an infinite strictly ascending chain of generated subvarieties.

The proof is based on an explicit sequence of finite algebras

$$(W_n)_{n\geq 1},$$

which we will show to belong to A-ROL. We then prove that for every $n \geq 2$, the algebra W_n is simple, and hence subdirectly irreducible, while $W_{n+1} \notin HS(W_2, \dots, W_n)$. Since A-ROL is congruence-distributive, Jónsson's theorem then implies that

$$W_{n+1} \notin V(W_2, \dots, W_n) \qquad \text{for all } n \geq 2,$$

which yields a strictly ascending chain of subvarieties of A-ROL.

5.1 The Family $(\mathbf{W}_n)_{n\geq 1}$

We begin by defining the algebras W_n and showing that each of them is an associative residuated ortholattice.

Definition 5.1. For each integer $n \geq 1$, define an algebra

$$\mathbf{W_n} = (W_n, \wedge, \vee, {}', 0, 1)$$

as follows.

Its underlying set is

$$W_n := \{0, 1\} \cup \{a_1, \dots, a_n\} \cup \{b_1, \dots, b_n\}.$$

We first define a partial order $\leq$ on W_n. The elements

$$0 < a_1 < \cdots < a_n < 1 \qquad \text{and} \qquad 0 < b_1 < \cdots < b_n < 1$$

form two chains from 0 to 1, and there are no other relations between the elements a_i and b_j. Equivalently, for all $i, j \in \{1, \dots, n\}$,

$$a_i \leq a_j \iff i \leq j, \qquad b_i \leq b_j \iff i \leq j,$$

while

$$a_i \not\leq b_j \qquad \text{and} \qquad b_j \not\leq a_i.$$

The lattice operations $\wedge$ and $\vee$ are then given by the meet and join with respect to this order. Explicitly, for all $i, j \in \{1, \ldots, n\}$,

$$a_i \wedge a_j = a_{\min(i,j)}, \qquad a_i \vee a_j = a_{\max(i,j)},$$

$$b_i \wedge b_j = b_{\min(i,j)}, \qquad b_i \vee b_j = b_{\max(i,j)},$$

and for elements from different branches,

$$a_i \wedge b_j = 0, \qquad a_i \vee b_j = 1.$$

Moreover, 0 is the least element and 1 is the greatest element, so

$$0 \wedge u = 0, \quad 0 \vee u = u, \quad 1 \wedge u = u, \quad 1 \vee u = 1 \qquad \text{for all } u \in W_n.$$

Finally, define the involution $'$ on W_n by

$$0' = 1, \qquad 1' = 0,$$

and, for each $i \in \{1, \ldots, n\}$,

$$a_i' = b_{n+1-i}, \qquad b_i' = a_{n+1-i}.$$

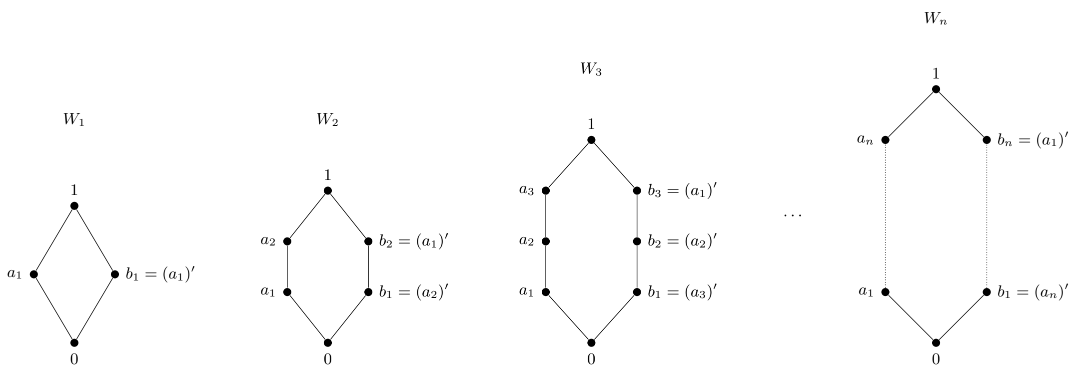


Figure 5.1 The first members of the sequence $(W_n)_{n\geq 1}$ and the general form of W_n.

We begin by verifying that the structures just defined are ortholattices.

Lemma 5.2. *For every $n \geq 1$, the algebra W_n is an ortholattice.*

Proof. First, $(W_n, \wedge, \vee, 0, 1)$ is a bounded lattice by construction, the meet and join of any two elements are explicitly given in the definition: for elements of the same branch, the minimum and maximum lie on that branch, while for elements lying on different branches the meet is 0 and the join is 1. Thus every pair of elements has both a meet and a join, and 0 and 1 are the least and greatest elements of W_n.

Next, $'$ is an involution. Clearly,

$$0'' = 0, \qquad 1'' = 1.$$

Moreover, for each $i \in \{1, \ldots, n\}$,

$$(a_i')' = (b_{n+1-i})' = a_{n+1-(n+1-i)} = a_i,$$

and analogously,

$$(b_i')' = (a_{n+1-i})' = b_{n+1-(n+1-i)} = b_i.$$

Hence $x'' = x$ for every $x \in W_n$.

We now show that $'$ is order-reversing. Suppose $a_i \leq a_j$. Then $i \leq j$, so $n+1-j \leq n+1-i$, and therefore

$$a_j' = b_{n+1-j} \leq b_{n+1-i} = a_i'.$$

The argument for $b_i \leq b_j$ is again analogous. Since there are no nontrivial comparabilities between elements from different branches, it follows that

$$x \leq y \implies y' \leq x'$$

for all $x, y \in W_n$.

Finally, we verify the orthocomplementation law. For 0 and 1 we have

$$0 \wedge 0' = 0 \wedge 1 = 0, \qquad 1 \wedge 1' = 1 \wedge 0 = 0.$$

If $i \in \{1, \ldots, n\}$, then $a_i' = b_{n+1-i}$, and these two elements lie on different branches. Hence

$$a_i \wedge a_i' = a_i \wedge b_{n+1-i} = 0.$$

Similarly,

$$b_i \wedge b_i' = b_i \wedge a_{n+1-i} = 0.$$

Thus $x \wedge x' = 0$ for every $x \in W_n$, and analogously $x \vee x' = 1$. Therefore W_n is an ortholattice. □

5.2 The Algebras $\mathbf{W}_n$ as Associative Residuated Ortholattices

We now show that the algebras W_n are residuated ortholattices and that they belong to A-ROL.

Lemma 5.3. *For every $n \geq 1$, the Sasaki product $*$ on W_n, defined by*

$$x * y := x \wedge (x' \vee y),$$

satisfies, for all $i, j \in \{1, \ldots, n\}$,

$$\begin{aligned} a_i * a_j = a_i, \qquad a_i * b_j = 0, \qquad b_i * a_j = 0,\\ b_i * b_j = b_i, \qquad 0 * u = 0, \qquad 1 * u = u. \end{aligned}$$

for every $u \in W_n$.

Proof. Let $i, j \in \{1, \dots, n\}$. Since $a_i' = b_{n+1-i}$, we have

$$a_i * a_j = a_i \wedge (a_i' \vee a_j) = a_i \wedge (b_{n+1-i} \vee a_j).$$

Now b_{n+1-i} and a_j lie on different branches, hence

$$b_{n+1-i} \vee a_j = 1.$$

Therefore

$$a_i * a_j = a_i \wedge 1 = a_i.$$

Then,

$$a_i * b_j = a_i \wedge (a_i' \vee b_j) = a_i \wedge (b_{n+1-i} \vee b_j).$$

Since b_{n+1-i} and b_j lie on the same branch, $b_{n+1-i} \vee b_j$ is again an element of the b-branch. Hence

$$a_i * b_j = 0,$$

because the meet of an a-element and a b-element is 0.

The proof of

$$b_i * a_j = 0, \qquad b_i * b_j = b_i$$

is analogous.

Finally, for every $u \in W_n$,

$$0 * u = 0 \wedge (0' \vee u) = 0 \wedge (1 \vee u) = 0,$$

and

$$1 * u = 1 \wedge (1' \vee u) = 1 \wedge (0 \vee u) = u.$$

□

Proposition 5.4. *For every $n \geq 1$, the ortholattice W_n is a residuated ortholattice.*

Proof. To show that each W_n is indeed a residuated ortholattice, we have to show that on every such lattice, residuation $\backslash$ of the Sasaki product $*$ is defined for all pairs of elements. We define a binary operation $\backslash$ on W_n by

$$0\backslash z = 1, \qquad 1\backslash z = z \qquad \text{for all } z \in W_n,$$

and, for each $i \in \{1, \dots, n\}$,

$$a_i\backslash z = \begin{cases} 1, & \text{if } a_i \leq z, \\ b_n, & \text{if } a_i \not\leq z, \end{cases} \qquad b_i\backslash z = \begin{cases} 1, & \text{if } b_i \leq z, \\ a_n, & \text{if } b_i \not\leq z. \end{cases}$$

We show that this operation is the residual of the Sasaki product on W_n. By the Remark 1.26 on residuation, it is enough to show that for all $x, z \in W_n$, the element $x\backslash z$ is the greatest $u \in W_n$ such that $x * u \leq z$.

First consider the case $a_i\backslash z$. Let $u \in W_n$. If $u = a_j$, then by Lemma 5.3,

$$a_i * u = a_i.$$

Hence

$$a_i * u \leq z \iff a_i \leq z.$$

If $u = b_j$, then

$$a_i * u = 0,$$

so

$$a_i * u \le z$$

always holds.

Therefore:

- if $a_i \le z$, then $a_i * u \le z$ for every $u \in W_n$, so the greatest such u (residual) is 1.
- if $a_i \not\le z$, then $a_i * u \le z$ holds precisely for $u = 0$ and for all u on the b-branch, and the greatest such element is b_n.

Thus $a_i \backslash z$ is the greatest element $u \in W_n$ such that $a_i * u \le z$.

The argument for $x = b_i$ is symmetric.

It remains to consider $0 \backslash z$ and $1 \backslash z$. By Lemma 5.3, for every $u \in W_n$,

$$0 * u = 0,$$

so $0 * u \le z$ always holds, and therefore the greatest such u is 1. Thus

$$0 \backslash z = 1.$$

Likewise,

$$1 * u = u,$$

so

$$1 * u \le z \iff u \le z.$$

Hence the greatest such u is z, and therefore

$$1 \backslash z = z.$$

We have proved that for all $x, z \in W_n$, the element $x \backslash z$ is the greatest element $u \in W_n$ such that

$$x * u \le z.$$

Consequently, for all $x, u, z \in W_n$,

$$x * u \le z \iff u \le x \backslash z.$$

Thus the Sasaki product on W_n is residuated, and W_n is a residuated ortholattice.
□

Now it remains to show that the algebras W_n are in fact associative residuated ortholattices. To this end, we first record the remaining values of the Sasaki product on W_n.

Lemma 5.5. *For every $n \ge 1$, the Sasaki product on W_n satisfies, for every $i \in \{1, \dots, n\}$,*

$$a_i * u = \begin{cases} a_i, & \textit{if } u \in \{1, a_1, \dots, a_n\}, \\ 0, & \textit{if } u \in \{0, b_1, \dots, b_n\}, \end{cases}$$

and

$$b_i * u = \begin{cases} b_i, & \text{if } u \in \{1, b_1, \ldots, b_n\}, \\ 0, & \text{if } u \in \{0, a_1, \ldots, a_n\}. \end{cases}$$

and

$$0 * u = 0, \qquad 1 * u = u \qquad \text{for every } u \in W_n.$$

Proof. By Lemma 5.3, for all $i, j \in \{1, \ldots, n\}$,

$$a_i * a_j = a_i, \qquad a_i * b_j = 0, \qquad b_i * a_j = 0, \qquad b_i * b_j = b_i,$$

and also

$$0 * u = 0, \qquad 1 * u = u \qquad \text{for every } u \in W_n.$$

It remains to determine the products with 0 and 1 on the right. Since W_n is an ortholattice by Lemma 5.2, the general identities for Sasaki product in bounded involutive lattices yield

$$x * 0 = x \wedge x' = 0, \qquad x * 1 = x \wedge (x' \vee 1) = x$$

for every $x \in W_n$.

Hence, for each i,

$$a_i * 0 = 0, \qquad a_i * 1 = a_i, \qquad b_i * 0 = 0, \qquad b_i * 1 = b_i.$$

Combining these with the values already obtained in Lemma 5.3 gives the stated description of $*$. □

Theorem 5.6. *For every $n \geq 1$, the algebra W_n is an associative residuated ortholattice. In particular,*

$$W_n \in \text{A-ROL}.$$

Proof. By Proposition 5.4, W_n is a residuated ortholattice. It remains to prove that the Sasaki product on W_n is associative.

Let $x, y, z \in W_n$. We show that

$$x * (y * z) = (x * y) * z.$$

If $x = 0$, then

$$x * (y * z) = 0 * (y * z) = 0, \qquad (x * y) * z = (0 * y) * z = 0 * z = 0.$$

Hence the equality holds.

If $x = 1$, then

$$x * (y * z) = 1 * (y * z) = y * z, \qquad (x * y) * z = (1 * y) * z = y * z.$$

So the equality again holds.

It remains to consider the cases where x lies on one of the two branches.

Case 1: $x = a_i$ for some $i \in \{1, \ldots, n\}$.

We then separate subcases based on y:

Subcase 1.1: $y = 0$. Then

$$y * z = 0 * z = 0,$$

so

$$x * (y * z) = a_i * 0 = 0.$$

On the other hand,

$$(x * y) * z = (a_i * 0) * z = 0 * z = 0.$$

Thus

$$a_i * (0 * z) = (a_i * 0) * z.$$

Subcase 1.2: $y = 1$. Then

$$y * z = 1 * z = z,$$

hence

$$x * (y * z) = a_i * z.$$

Also,

$$(x * y) * z = (a_i * 1) * z = a_i * z.$$

So

$$a_i * (1 * z) = (a_i * 1) * z.$$

Subcase 1.3: $y = a_j$ *for some* $j \in \{1, \ldots, n\}$. By Lemma 5.5,

$$a_j * z = \begin{cases} a_j, & \text{if } z \in \{1, a_1, \ldots, a_n\}, \\ 0, & \text{if } z \in \{0, b_1, \ldots, b_n\}. \end{cases}$$

Therefore

$$a_i * (a_j * z) = \begin{cases} a_i * a_j = a_i, & \text{if } z \in \{1, a_1, \ldots, a_n\}, \\ a_i * 0 = 0, & \text{if } z \in \{0, b_1, \ldots, b_n\}. \end{cases}$$

On the other hand,

$$(a_i * a_j) * z = a_i * z,$$

and again by Lemma 5.5,

$$a_i * z = \begin{cases} a_i, & \text{if } z \in \{1, a_1, \ldots, a_n\}, \\ 0, & \text{if } z \in \{0, b_1, \ldots, b_n\}. \end{cases}$$

Hence

$$a_i * (a_j * z) = (a_i * a_j) * z.$$

Subcase 1.4: $y = b_j$ *for some* $j \in \{1, \ldots, n\}$. By Lemma 5.5,

$$b_j * z = \begin{cases} b_j, & \text{if } z \in \{1, b_1, \ldots, b_n\}, \\ 0, & \text{if } z \in \{0, a_1, \ldots, a_n\}. \end{cases}$$

In either case,

$$a_i * (b_j * z) = 0,$$

because $a_i * b_j = 0$ and $a_i * 0 = 0$. On the other hand,

$$(a_i * b_j) * z = 0 * z = 0.$$

Thus

$$a_i * (b_j * z) = (a_i * b_j) * z.$$

This completes the proof when $x = a_i$.

Case 2: $x = b_i$ for some $i \in \{1, \dots, n\}$.

This is entirely symmetric to Case 1, using the second part of Lemma 5.5. Hence

$$b_i * (y * z) = (b_i * y) * z \qquad \text{for all } y, z \in W_n.$$

We have exhausted all possibilities for $x \in W_n$. Therefore

$$x * (y * z) = (x * y) * z \qquad \text{for all } x, y, z \in W_n.$$

Thus the Sasaki product on W_n is associative.

Hence W_n is an associative residuated ortholattice, that is,

$$W_n \in \text{A-ROL}.$$

□

An important property of the family $(W_n)_{n \geq 1}$ is that the product on each W_n is branch-sensitive when multiplying by the Sasaki product: multiplying along the same branch returns the left coordinate, while multiplying across branches collapses to 0. This makes the associativity proof manageable by a case distinction, but at the same time keeps the algebras far from trivial and ultimately allows them to separate subvarieties of A-ROL.

The first two members have the following additional properties. W_1 is the four-element Boolean algebra with underlying set

$$W_1 = \{0, 1, a_1, b_1\},$$

and with $a_1' = b_1$ and $b_1' = a_1$, while $a_1 \wedge b_1 = 0$ and $a_1 \vee b_1 = 1$. Hence $W_1 \in \text{BA} \subseteq \text{A-ROL}$.

W_2 is the first non-Boolean member of the sequence. Its ortholattice reduct is isomorphic to the six-element ortholattice B_6 (benzene). By Lemma 3.9, it follows that W_2 is not orthomodular. Since $W_2 \in \text{A-ROL}$ by Theorem 5.6, it follows that W_2 gives an explicit witness for

$$\text{A-ROL} \nsubseteq \text{OML}.$$

In particular, W_2 witnesses that not every residuated ortholattice is orthomodular.

5.3 Congruence Properties of $\mathbf{W}_n$

We now turn to the general universal-algebraic part of the argument. Our goal is to show that the variety A-ROL has infinitely many subvarieties using Jónsson's theorem for finitely generated congruence-distributive varieties. For that we first need to show that A-ROL is congruence-distributive.

Lemma 5.7. *The variety* A-ROL *is congruence-distributive.*

Proof. Let
$$\mathbf{A} = (A, \wedge, \vee, ', \backslash, 0, 1) \in \text{A-ROL}.$$
Since $(A, \wedge, \vee)$ is a lattice, its congruence lattice
$$\mathrm{Con}(A, \wedge, \vee)$$
is distributive by the Funayama–Nakayama theorem, which states that the congruence lattice of every lattice is distributive [8]. Every congruence of $\mathbf{A}$ is, in particular, a congruence of its lattice reduct, so $\mathrm{Con}(\mathbf{A})$ is a sublattice of $\mathrm{Con}(A, \wedge, \vee)$. Since every sublattice of a distributive lattice is distributive, it follows that $\mathrm{Con}(\mathbf{A})$ is distributive. Therefore A-ROL is congruence-distributive. $\square$

We now analyze the congruences of the algebras W_n. We will show that for all $n \geq 2$, these algebras are *simple*.

Proposition 5.8. *For every $n \geq 2$, the algebra*
$$(W_n, \wedge, \vee, ', \backslash, 0, 1)$$
is simple, i.e.
$$\mathrm{Con}(W_n) = \{\Delta_{W_n}, \nabla_{W_n}\}.$$

Proof. Fix $n \geq 2$, and let $\theta \in \mathrm{Con}(W_n)$ with
$$\theta \neq \Delta_{W_n}.$$
We prove that
$$\theta = \nabla_{W_n}.$$

Since $\theta \neq \Delta_{W_n}$, there exist distinct elements $x, y \in W_n$ such that
$$x \; \theta \; y.$$

We distinguish cases.

Case 1: x and y lie on the same branch.

Assume first that
$$x = a_i, \qquad y = a_j, \qquad i \neq j.$$
Without loss of generality, let $i < j$. Since θ is compatible with $\backslash$, from
$$a_i \; \theta \; a_j$$
we obtain
$$a_i \backslash a_j \; \theta \; a_j \backslash a_i.$$
Because $a_i \leq a_j$ but $a_j \not\leq a_i$, the definition of $\backslash$ gives
$$a_i \backslash a_j = 1, \qquad a_j \backslash a_i = b_n.$$
Hence
$$1 \; \theta \; b_n.$$

Now apply the unary term $a_n \wedge x$ to the relation $1 \; \theta \; b_n$. Since
$$a_n \wedge 1 = a_n, \qquad a_n \wedge b_n = 0,$$

we obtain

$$a_n \,\theta\, 0.$$

Applying the involution $'$ to $a_i \,\theta\, a_j$, we also get

$$b_{n+1-i} \,\theta\, b_{n+1-j}.$$

Since $i < j$, we have $n+1-j < n+1-i$, and therefore

$$b_{n+1-j} \backslash b_{n+1-i} = 1, \qquad b_{n+1-i} \backslash b_{n+1-j} = a_n.$$

Thus

$$1 \,\theta\, a_n.$$

Together with $a_n \,\theta\, 0$, this yields

$$1 \,\theta\, 0.$$

Therefore $\theta = \nabla_{W_n}$.

The case

$$x = b_i, \qquad y = b_j, \qquad i \neq j,$$

is symmetric.

Case 2: x and y lie on different branches.

Assume

$$x = a_i, \qquad y = b_j$$

for some $i, j \in \{1, \ldots, n\}$. Applying the unary term $a_i \wedge -$ to

$$a_i \,\theta\, b_j,$$

we get

$$a_i = a_i \wedge a_i \,\theta\, a_i \wedge b_j = 0.$$

Thus

$$a_i \,\theta\, 0.$$

Now let $k \in \{1, \ldots, n\}$. By Lemma 5.3,

$$a_k * a_i = a_k, \qquad a_k * 0 = 0.$$

Since θ is compatible with $*$, from $a_i \,\theta\, 0$ we obtain

$$a_k = a_k * a_i \,\theta\, a_k * 0 = 0.$$

Thus every a_k is θ-equivalent to 0. In particular,

$$a_1 \,\theta\, a_2.$$

Since $n \geq 2$, Case 1 applies, and therefore

$$\theta = \nabla_{W_n}.$$

The case $x = b_j$, $y = a_i$ is symmetric.

Case 3: one of x, y is 0 or 1.

Assume first that

$$0 \;\theta\; a_i$$

for some $i \in \{1, \ldots, n\}$. Then for every $k \in \{1, \ldots, n\}$,

$$a_k = a_k * a_i \;\theta\; a_k * 0 = 0.$$

Hence in particular

$$a_1 \;\theta\; a_2.$$

Since $n \geq 2$, Case 1 applies, and therefore

$$\theta = \nabla_{W_n}.$$

The case

$$0 \;\theta\; b_i$$

is symmetric.

If

$$1 \;\theta\; a_i \quad \text{or} \quad 1 \;\theta\; b_i,$$

apply the involution $'$ and reduce to one of the previous subcases.

Finally, if

$$0 \;\theta\; 1,$$

then clearly

$$\theta = \nabla_{W_n}.$$

These cases exhaust all possibilities for distinct $x, y \in W_n$. Hence every congruence $\theta \in \mathrm{Con}(W_n)$ satisfying $\theta \neq \Delta_{W_n}$ must be equal to ∇_{W_n}. Therefore

$$\mathrm{Con}(W_n) = \{\Delta_{W_n}, \nabla_{W_n}\},$$

so W_n is simple. □

Lemma 5.9. *Every simple algebra is subdirectly irreducible.*

Proof. If $\mathbf{A}$ is simple, then

$$\mathrm{Con}(\mathbf{A}) = \{\Delta_A, \nabla_A\}.$$

Hence ∇_A is the least nontrivial congruence on $\mathbf{A}$, so $\mathbf{A}$ is subdirectly irreducible. □

Corollary 5.10. *For every $n \geq 2$, the algebra W_n is subdirectly irreducible.*

Proof. Since W_n is simple for every $n \geq 2$, Lemma 5.9 implies that W_n is subdirectly irreducible for every $n \geq 2$. □

5.4 Jónsson's Theorem and the Infinite Chain

We have now established all the properties of the sequence $(W_n)_{n\geq 1}$ needed for the final argument. To finally prove that the variety A-ROL has infinitely many subvarieties, we will need the following definitions and theorems:

Definition 5.11. Let $\mathcal{K}$ be a class of algebras of a fixed language.

We denote by

$$I(\mathcal{K}), \qquad S(\mathcal{K}), \qquad H(\mathcal{K}), \qquad P(\mathcal{K}), \qquad P_U(\mathcal{K})$$

the classes of all algebras that are, respectively, isomorphic copies, subalgebras, homomorphic images, direct products, and ultraproducts of members of $\mathcal{K}$.

We also write

$$HS(\mathcal{K}) := H(S(\mathcal{K}))$$

and

$$HSP_U(\mathcal{K}) := H\Big(S(P_U(\mathcal{K}))\Big).$$

A nonempty class $\mathcal{V}$ of algebras of the given language is called a *variety* if it is closed under homomorphic images, subalgebras, and direct products.

We denote by

$$V(\mathcal{K})$$

the smallest variety containing $\mathcal{K}$. Equivalently,

$$V(\mathcal{K}) = HSP(\mathcal{K}).$$

Theorem 5.12 (Jónsson's Theorem)**.** *Let $\mathcal{V}$ be a congruence-distributive variety and let $\mathcal{K} \subseteq \mathcal{V}$. If $\mathbf{A}$ is subdirectly irreducible and*

$$\mathbf{A} \in V(\mathcal{K}),$$

then

$$\mathbf{A} \in HSP_U(\mathcal{K}).$$

Proof. See [4, Theorem IV.6.8, p. 165]. □

In our case, the generating classes $K_n = \{W_2, \dots, W_n\}$ are finite sets of finite algebras. In that situation, the role of ultraproducts occurring in Jónsson's theorem is simplified by the following lemma:

Lemma 5.13. *Let $\{A_i : i \in I\}$ be a family of algebras such that the set $\{A_i : i \in I\}$ is finite and each A_i is finite. Let U be an ultrafilter on I. Then the ultraproduct*

$$\prod_{i\in I} A_i/U$$

is isomorphic to one of the algebras occurring among the A_i.

Proof. See [4, Lemma IV.6.5]. □

Thus, when $\mathcal{K}$ is a finite set of finite algebras, every member of $P_U(\mathcal{K})$ is isomorphic to a member of $\mathcal{K}$. Equivalently,

$$P_U(\mathcal{K}) \subseteq I(\mathcal{K}).$$

Combining this with Jónsson's theorem yields the finite version that we need.

Corollary 5.14 (Jónsson, finite case)**.** *Let $\mathcal{K}$ be a finite set of finite algebras. If $V(\mathcal{K})$ is congruence-distributive, then every subdirectly irreducible algebra in $V(\mathcal{K})$ belongs to $HS(\mathcal{K})$.*

Proof. This is a direct consequence of Theorem 5.12 and Lemma 5.13 for finite sets of finite algebras. □

We now apply this finite consequence of Jónsson's theorem to the sequence of algebras W_n. The next lemma shows that W_{n+1} cannot occur as a homomorphic image of a subalgebra of one of the earlier algebras $W_2, \dots, W_n$.

Lemma 5.15. *For every $n \geq 2$,*

$$W_{n+1} \notin HS(W_2, \dots, W_n).$$

Proof. Fix $n \geq 2$, and suppose towards a contradiction that

$$W_{n+1} \in HS(W_2, \dots, W_n).$$

Then, by definition of HS, there exist some $i \in \{2, \dots, n\}$, a subalgebra

$$B \leq W_i,$$

and a surjective homomorphism

$$B \twoheadrightarrow W_{n+1}.$$

Hence

$$|W_{n+1}| \leq |B| \leq |W_i|.$$

Since $i \leq n$, we have

$$|W_i| = 2i + 2 \leq 2n + 2.$$

On the other hand,

$$|W_{n+1}| = 2(n+1) + 2 = 2n + 4.$$

Thus

$$|W_{n+1}| > |W_i|,$$

a contradiction. Therefore

$$W_{n+1} \notin HS(W_2, \dots, W_n).$$

□

We are now ready to prove the main result of this chapter.

Theorem 5.16. *The variety* A-ROL *has infinitely many subvarieties.*

Proof. For each $n \geq 2$, let

$$K_n := \{W_2, \dots, W_n\}.$$

We claim that

$$W_{n+1} \notin V(K_n) \qquad \text{for every } n \geq 2.$$

Fix $n \geq 2$, and suppose towards a contradiction that

$$W_{n+1} \in V(K_n).$$

Since $V(K_n) \subseteq$ A-ROL, Lemma 5.7 implies that $V(K_n)$ is congruence-distributive. The set K_n is finite, and each algebra in K_n is finite. Moreover, by Corollary 5.10, the algebra W_{n+1} is subdirectly irreducible. Therefore Corollary 5.14 applies and yields

$$W_{n+1} \in HS(K_n).$$

This contradicts Lemma 5.15. Hence

$$W_{n+1} \notin V(K_n) \qquad \text{for all } n \geq 2.$$

Now $K_n \subseteq K_{n+1}$, so

$$V(K_n) \subseteq V(K_{n+1}).$$

Also,

$$W_{n+1} \in K_{n+1} \subseteq V(K_{n+1}).$$

At the same time, we have previously proved that $W_{n+1} \notin V(K_n)$. Thus

$$V(K_n) \neq V(K_{n+1}).$$

and therefore

$$V(K_n) \subsetneq V(K_{n+1}) \qquad \text{for every } n \geq 2.$$

From this we can derive that

$$V(W_2) \subsetneq V(W_2, W_3) \subsetneq V(W_2, W_3, W_4) \subsetneq \cdots$$

is a strictly ascending chain of subvarieties of A-ROL. Consequently, A-ROL has infinitely many subvarieties. $\square$

The argument has the following structure: First, congruence-distributivity allows us to use Jónsson's theorem. Second, because the generating sets K_n are finite sets of finite algebras, the ultraproducts in Jónsson's theorem reduce to members of K_n itself, so subdirectly irreducible members of $V(K_n)$ must already lie in $HS(K_n)$. Finally, the cardinality argument in Lemma 5.15 shows that W_{n+1} does not belong to $HS(W_2, \ldots, W_n)$, and hence not to $V(W_2, \ldots, W_n)$. This separates each W_{n+1} from the variety generated by its predecessors and yields the infinite strictly ascending chain of subvarieties.

Conclusion

This thesis studied residuated ortholattices, with particular emphasis on their associative subvariety.

The thesis first situated the classes ROL and A-ROL within the surrounding lattice-theoretic landscape. In particular, it showed that ROL is a variety properly extending the class OML of orthomodular lattices, that A-ROL is a proper subvariety of ROL, that A-ROL and OML are incomparable under inclusion, and that Boolean algebras form a proper subclass of A-ROL.

The thesis then analyzed the effect of associativity on the Sasaki product. It was shown that in every residuated ortholattice the Sasaki product is idempotent and satisfies the left regular identity, so that in A-ROL the reduct $(A, *)$ is a left regular band. It was further proved that adding the identity $lr2$ forces collapse to Boolean algebras. Equivalently, in the pure language of the Sasaki product, every proper equational strengthening of A-ROL is already Boolean.

The main result of the thesis is that the variety A-ROL has infinitely many subvarieties. This was proved by constructing an explicit sequence $(W_n)_{n \geq 1}$ of finite algebras, showing that each W_n belongs to A-ROL, proving simplicity of W_n for every $n \geq 2$, and applying congruence-distributivity together with Jónsson's theorem to obtain a strictly ascending chain of varieties

$$V(W_2) \subsetneq V(W_2, W_3) \subsetneq V(W_2, W_3, W_4) \subsetneq \cdots .$$

These results show that associativity does not collapse residuated ortholattices to a mathematically trivial class. The associative subvariety admits a nontrivial internal structure, although its proper subvarieties cannot be distinguished by equations in the pure language of the Sasaki product alone.

Two natural directions for further research suggest themselves.

The first is to continue the study of A-ROL from the perspective of universal algebra. A natural next step would be to obtain a more detailed description of the lattice of subvarieties of A-ROL, for example by identifying equations that separate the varieties generated by the algebras W_n, and whether the variety generated by the family $(W_n)_{n \geq 1}$ coincides with all of A-ROL.

The second is to clarify the logical significance of A-ROL and of its subvarieties.

In this sense, A-ROL remains closely connected to the orthomodular setting through the Sasaki product, while being algebraically rich enough to support a nontrivial theory of subvarieties. This makes it a promising object for further study.

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